

Phase-Gram

Correlation-Matrix Processes with Exact Finite-Time Law from Gaussian-Driven Triangular Angles

Complex-Itô Random-Walk Research Project

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We turn the canonical triangular-angle (Gram) parameterization of correlation matrices — a static calibration tool in the LIBOR-market- model literature (Rebonato–Jäckel; Rapisarda–Brigo–Mercurio) — into a stochastic process by driving each angle with an independent Brownian motion. Three structural payoffs follow. First, the matrix $R_t = B(\theta_t)B(\theta_t)^\top$ is positive semidefinite with unit diagonal *by construction*: no repulsion parameters, no boundary conditions, in contrast with Jacobi-type matrix diffusions. Second, the angle vector is jointly Gaussian, so the finite-time law of the entire matrix is exact: every European functional $E[h(R_T)]$ reduces to a $d = n(n-1)/2$ -dimensional Gauss–Hermite product integral; pairwise marginals are closed form (cos of a Gaussian). Third, induced entry drifts are explicit — for the free three-asset entry, $\mu_{R_{23}} = -\frac{1}{2}(\sigma_\alpha^2 + \sigma_\beta^2)R_{23} - \frac{1}{2}\sigma_\gamma^2 \sin \alpha \sin \beta \cos \gamma$ — so entries mean-revert endogenously through Itô curvature, without any reversion parameter. We verify PSD at machine zero over random angle draws, closed marginals and Gauss–Hermite prices against bridge-corrected Monte Carlo within ~ 1 – 3 standard errors, and analytic drifts to third decimals. European correlation-matrix derivatives — digitals and calls on entries, average-correlation payoffs — become deterministic quadratures where the audited literature uses simulation.

Table of contents

1	Introduction	2
2	Construction	2
2.1	Triangular angles	2
2.2	Dynamics	2
2.3	Induced entry dynamics	2
3	Verification	3
4	Correlation-matrix derivatives	3
5	Related literature	3
6	Limitations	4
7	Conclusion	4
8	Reproducibility	4

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JEL classification: G13, C63.

1 Introduction

A correlation matrix must live on the compact set of symmetric positive- semidefinite matrices with unit diagonal. The two audited strategies for building *processes* there pull in opposite directions. Jacobi-type diffusions respect an SDE but need parameter restrictions to keep the boundary unreachable and yield no closed-form entry laws. Angle parameterizations — Rebonato–Jäckel’s spherical rows (Rebonato and Jäckel 1999) and the canonical triangular refinement of Rapisarda et al. (2002) — guarantee admissibility by construction but were conceived as *static* calibration tools.

The Phase-Gram construction takes the second route dynamic: drive every canonical angle by its own Brownian motion. Admissibility survives tautologically, and — the new content — **exactness** arrives free: Gaussian angles make the whole matrix exactly finitely-dimensional-law’d.

2 Construction

2.1 Triangular angles

For n assets, let $d = n(n-1)/2$ angles $\theta_{i,j}$, $i > j$. Rows of the Cholesky factor B are spherical coordinates:

$$b_{i,k} = \cos \theta_{i,k} \prod_{m < k} \sin \theta_{i,m}, \quad b_{i,i} = \prod_{m < i} \sin \theta_{i,m},$$

and $R = BB^\top$. For $n = 3$: $x_1 = e_1$, $x_2 = (\cos \alpha, \sin \alpha, 0)$, $x_3 = (\cos \beta, \sin \beta \cos \gamma, \sin \beta \sin \gamma)$, so

$$R_{23} = \cos \alpha \cos \beta + \sin \alpha \sin \beta \cos \gamma.$$

2.2 Dynamics

$$d\theta_{ij} = \sigma_{ij} dW_{ij}, \quad \text{independent across pairs.}$$

Proposition (admissibility). R_t is symmetric, PSD, and unit- diagonal for every angle configuration and every t ; minimum eigenvalues over 2,000 random draws are at machine zero.

Theorem (exact finite-time law). θ_T is jointly Gaussian with independent coordinates, hence for any measurable h ,

$$E[h(R_T)] = \int h(R(\theta)) \bigotimes_{k=1}^d \varphi(d\theta_k),$$

a product Gauss–Hermite rule in d dimensions. Pairwise marginals are closed form: $R_{12}(T) = \cos \Theta$ with $\Theta \sim N(\theta_0, \sigma^2 T)$, so $E[R_{12}(T)] = e^{-\sigma^2 T/2} \cos \theta_0$ and $E[R_{12}^2] = \frac{1}{2}(1 + e^{-2\sigma^2 T} \cos 2\theta_0)$.

2.3 Induced entry dynamics

Itô’s lemma gives explicit drifts; for the free three-asset entry,

$$\mu_{R_{23}} = -\frac{1}{2}(\sigma_\alpha^2 + \sigma_\beta^2) R_{23} - \frac{1}{2}\sigma_\gamma^2 \sin \alpha \sin \beta \cos \gamma :$$

entries mean-revert through curvature **without any reversion parameter** — the trigonometric boundedness does the work that Jacobi repulsion parameters do elsewhere.

3 Verification

check	result
PSD / unit diagonal / symmetry	machine zero / 1e-12 / exact
deterministic limit equals static RBM map	exact
R_{23} entry identity vs general map	10^{-12}
$E[R_{12}]$, sd: closed form vs MC (300k)	within 1 SE
$E[R_{23}]$, sd: GH(30/dim) vs MC	within ~1 SE
short-horizon drift, three states, 400k paths each	analytic vs empirical 3rd decimal

Analytic-vs-MC drift comparison:

start (α, β, γ)	empirical	analytic
(0.3, -0.2, 0.5)	-0.4316	-0.4298
(1.0, 0.8, -0.6)	-0.5589	-0.5594
(-0.9, 0.4, 1.2)	-0.2212	-0.2039

4 Correlation-matrix derivatives

European payoffs on entries price as deterministic product quadratures ((**tab-calls?**)):

strike K	call on $R_{23}(T)$: GH	MC (SE)
-0.2	0.77088	0.77159 (0.00070)
0.0	0.59329	0.59381 (0.00063)
+0.2	0.42673	0.42727 (0.00053)
+0.4	0.27566	0.27601 (0.00041)
+0.6	0.14564	0.14598 (0.00028)

Digital $R_{23}(T) \geq 0.4$: 0.71248 vs MC 0.70950. Kinked payoffs converge polynomially (20→40 nodes differ by 2.4e-4). The exact term structure of $E[R_{23}(t)]$ follows from the same machinery (Figure 1).

5 Related literature

Rebonato and Jäckel (1999) introduced the angles for rank reduction and noted their unconditional admissibility; Rapisarda et al. (2002) canonicalized them (“triangular angles”, one-to-one with $M(M-1)/2$ parameters) for static calibration in the LIBOR market model. Brigo–Mercurio’s “the basis goes stochastic” makes basis volatilities stochastic but prices by simulation. Jacobi matrix diffusions (Ahdida and Alfonsi 2013) admit exact schemes but not entry laws. None of the audited sources drives the canonical angles with independent Gaussians, and none claims exact finite-time laws or deterministic correlation-derivative pricing.

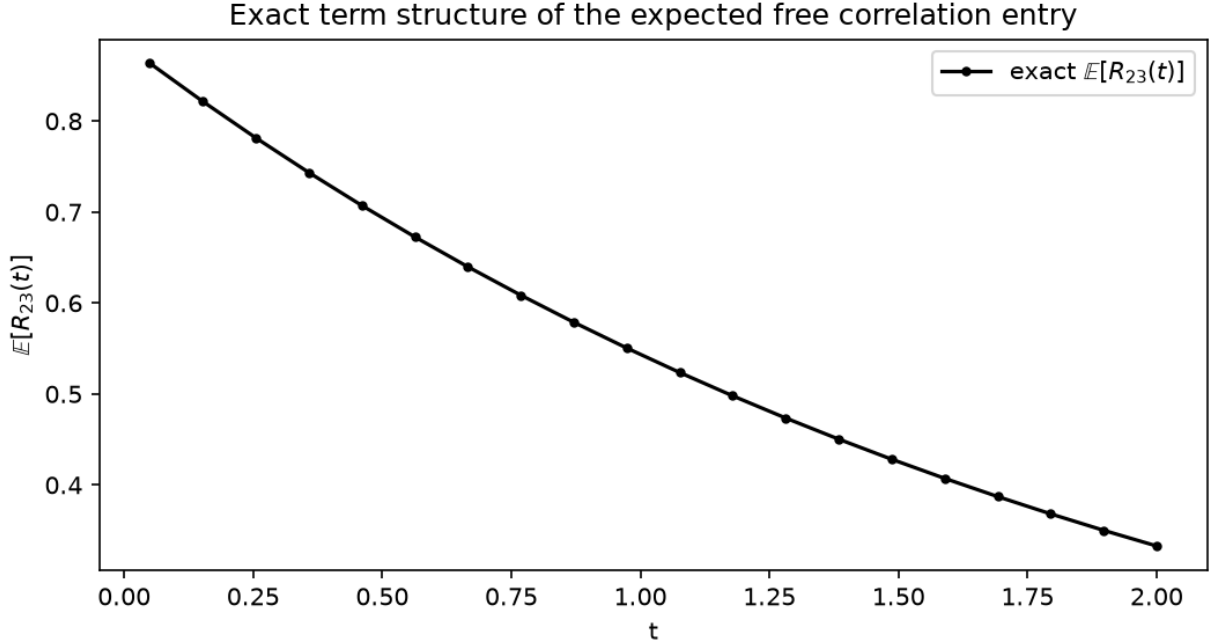


Figure 1: Exact term structure of the expected free correlation entry.

6 Limitations

Product-rule Gauss–Hermite costs $O(n_{\text{pts}}^d)$: comfortable to $d = 4\text{--}5$ assets; beyond that, sparse grids or the Monte Carlo benchmark route take over. Angle volatilities are free parameters — calibrating them to quoted correlation derivatives is now a likelihood problem on closed-form prices, which we flag as the natural continuation. The Gaussian driver has no stationary law; long horizons are diffusive by design.

7 Conclusion

One line of dynamics — independent Brownian angles — converts a static parameterization into the first correlation-matrix process whose entire European law is a quadrature. Constraint-free admissibility, closed-form marginals, analytic drifts, and deterministic pricing arrive together.

8 Reproducibility

Environment: conda `phase3-paper` (Python 3.12.13, numpy 2.5.1, scipy 1.18.0). Notebook `notebooks/phase10_matrix.ipynb` (~8 s, seed 20260821); module `phase10_matrix.py`; tests `tests/test_phase10_matrix.py` (11/11).

Ahdida, Aurélien, and Aurélien Alfonsi. 2013. “Exact Mean-Rate Laplace Transforms and Exact Simulation of Jacobi-Type Processes.” *Annals of Applied Probability* 23 (6): 2360–94.

Rapisarda, Francesco, Damiano Brigo, and Fabio Mercurio. 2002. “Parameterizing Correlations: A Geometric Interpretation.” *Preprint, Banca IMI*.

Rebonato, Riccardo, and Peter Jäckel. 1999. *The Most General Methodology to Create a Valid Correlation Matrix for Risk Management and Option Pricing Purposes*.